

Beyond Backside Power: Backside Signal Routing as Technology Booster for Standard-Cell Scaling

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ABSTRACT Advances in process technology enabling backside metals (BSMs) and contacts offer new design–technology co-optimization (DTCO) opportunities to further enhance power, performance, and area gains (PPA) in sub-3-nm nodes. This work exploits backside (BS) contact technology within standard cells to extend both signal and clock routing to BSM layers, enabling standard-cell height reduction options. We design electrically equivalent (EEQ) standard cells with multiple layout variants based on front versus BS pin access, achieving a 2-M0-Track height reduction in 3-nm gate-all-around field-effect transistor (GAAFET) technology. Experimental evaluation across representative industrial benchmarks—including high-performance CPUs, GPUs, and general-purpose systems-on-chip (SoCs) demonstrates significant benefits. Cell height reduction delivers up to 35% area savings and 10%–15% total power reduction for GPU and GP-SoC designs. For high-performance CPUs, maximum performance improves by 15% at iso-power compared to backside power with buried power rails (BSBPR). Incorporating BS signal routing with cell height reduction also reduces worst case IR drop by 32% relative to BSBPR. These results show that BS clock (BSCLK) and signal routing represent the next phase of technology innovation beyond BS power delivery, enabling continued standard-cell scaling, improved intracell and intercell routability, and generational PPA gains while maintaining similar core transistor geometries in sub-3-nm technologies.

INDEX TERMS Backside (BS) contacts, BS interconnects, BS signal, deep learning accelerators, design–technology co-optimization (DTCO), graphics processing units, high-performance compute.

I. INTRODUCTION

BACKSIDE (BS). Limitations with nano-through silicon via (nano-TSV) scaling and wafer thinning abilities previously restricted BS metals (BSMs) for power delivery only [3], [4]. Recent advances in BSM processing have shown the ability to connect gate, source/drain contacts directly to BSMs using ultrafine-pitch back-side contacts (BSCs) [5], [6], which have opened up a new design space as shown in Fig. 1.

However, further dimensional scaling and standard-cell height reduction result in fewer metal tracks available per

cell, worsening the routing congestion and restricting the standard-cell pin access. In contrast to device scaling, slower interconnect scaling limits interconnect pitches of lower back-end-of-line (BEOL) layers metals on the frontside (FS), exacerbating the standard-cell pin access problems. This limits the frequency gains and makes the power, area versus performance tradeoff more acute [7]. While BSCs for power delivery have shown ability to scale standard-cell height previously [8], [9], no prior work has leveraged BSC for standard-cell routing and height reduction. This work proposes a standard-cell design methodology incorporating BS

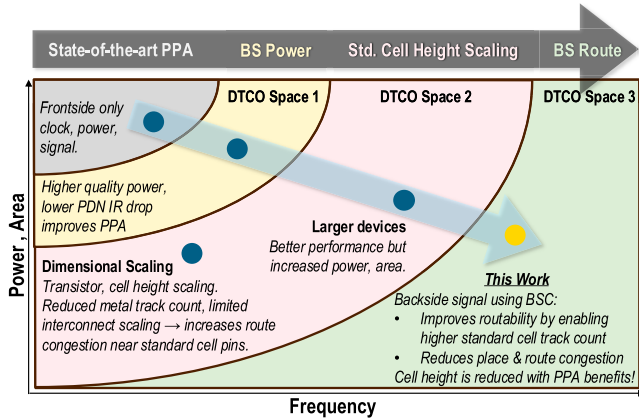


FIGURE 1. Dimensional scaling reduces the number of metal tracks available for routing per standard cell, intensifying routing congestion near cell I/O pins. Incorporating BS interconnects and contacts provides additional routing resources on the BS for signals, clocks, and power and increases the total metal track count per cell (FS + BS), facilitating further cell height reduction, thereby enhancing PPA metrics.

interconnects and contacts for signal routing, to assess PPA benefits at both the standard cell and IP levels. Through design–technology co-optimization (DTCO), this study seeks to improve metal track availability within standard cells and to address routing congestion challenges associated with reduced standard-cell heights in future technology nodes. The research is structured around targeted investigations aimed at achieving the following objectives.

- 1) We explore electrically equivalent (EEQ) standard-cell layouts (logically identical layout variations for the same standard cell/group of cells) to enable BS pin access with BSC [5]. This approach provides an alternative to overcome BSC/TSV pitch scaling limitations caused by overlay and misalignment errors [8], [10]. By moving I/O pin access to the BS via BSC, both FS and BSM tracks become available for routing. This increases track count per cell, enabling standard-cell height reduction and improving PPA, without requiring transistor scaling (yellow circle in Fig. 1).
- 2) DTCO of clock and signal routing on BSM layers with direct standard-cell pin access through BSC. We analyze the impact of reduced height cells on placement and routing congestion, area utilization, performance, and power consumption on a range of industrial representative benchmark test cases.
- 3) By analyzing area, performance scaling, we identify practical trends and tradeoffs in PPA metrics as a function of technology and physical design parameters, providing key insights for the development of BSM stacks in sub-3-nm nodes.

Three high-performance compute benchmarks: 1) *ibex*, a 32-bit RISC-V processor [11]; 2) data encryption standard IP core optimized for performance *des_perf* [12]; and

3) representative processing elements of GPU benchmark like convolution, multiply, and accumulate (CMAC) unit of open-source NVIDIA deep learning accelerator NVDLA [13], are used for exploring performance and dynamic power scaling trends with BS interconnects. For exploring static/total power and area scaling trends, we use three additional general-purpose system-on-chip (SoC) benchmarks: 1) memory controllers *dma_ctrl*; 2) communication IP *wb_conmax*; and 3) encryption cores *aes_core* [12], which are either cell and/or wire dominant along with the high-performance designs.

The remaining sections of this article are organized as follows. Section II reviews prior work on BS interconnects, DTCO approaches, and recent trends in standard-cell height scaling with BS routing. Section III outlines the technological modeling framework, including BS interconnect modeling and standard-cell design with BSMs for intracell routing and pin access. Section IV details the benchmark setup and ASIC design methodology, while Section V presents the experimental validation of cell height reduction. Finally, Section VI summarizes the key findings and conclusions.

II. RELATED WORK

A. BS INTERCONNECTS

Technological advances in semiconductor space have shown BS interconnects as a key enabler for achieving generational technological gains for continued standard-cell scaling and performance improvements in both 2-D and 3-D SoC designs. Backside power with buried power rails (BSBPR) significantly enhances both delay and power efficiency, with performance improvements of up to 5%–10% and energy efficiency of up to 20% for logic cells compared to traditional FS routing [14], [15]. Integrating TSVs into standard cells enables BS clock (BSCLK) routing, reducing clock power by 32% [16]. Such innovations are particularly impactful at technology nodes of 3 nm and below, where the decoupling of power and signal paths on opposite sides of wafer allows for reduced cell heights and increased routing flexibility. This facilitates further area reduction and performance improvement. Recent works have also explored thermal effects, manufacturing and integration challenges of BSMs, highlighting their potential to address scaling bottlenecks in 2-D, 2.5-D, and 3-D systems [17]. Most recent works in [5] and [6] have explored direct contact to the transistor source/drain from BS using BSCs. For beyond nanosheet technologies like ribbonFET, Park et al. [18] have shown direct access to the gate of transistor from BS.

B. DESIGN–TECHNOLOGY CO-OPTIMIZATION

DTCO has emerged as a critical methodology for maximizing benefits from BS metallization technology. In the case of BSMs, DTCO enables tuning of optimal front-backside vias (BVs) and aspect ratio of BSM stack to maximize performance benefit while adhering to manufacturing constraints [17], [19]. Machine learning-enabled DTCO

flows have further accelerated path-finding and routability assessments [20], resulting in rapid evaluation of standard-cell architectures. Reliability and variability-aware modeling through DTCO also enable bridging the gap between simulation and manufacturing [21].

C. STANDARD-CELL HEIGHT SCALING WITH BSC

This technology boosting measure has demonstrated substantial improvements in power delivery and area efficiency for advanced CMOS nodes. Recent works in enabling BSCs for BS power either using nano-TSV and BPR [22] or using direct BSCs [5], [6] allow for higher dense standard-cell libraries, with area gains of up to 20% and delay improvements of 10% in nanosheet architectures [9], [23]. BSCs/metals further enhance IR-drop reduction and support aggressive track height compression [24]. Such advances collectively highlight the importance of exploring BSMs/contacts in enabling continued standard-cell scaling, improved intracell and intercell routability, and generational PPA gains for sub-3-nm technology nodes [22], [23].

III. TECHNOLOGY MODELING

This section elaborates on BS interconnect modeling to achieve clock, signal routing on BS, and design of EEQ standard cells for enabling BS pin access. For technology modeling, we used an open-source 3-nm nanosheet-based gate-all-around field-effect transistor (GAAFET) process design kit (PDK) [25] and modified the metal stack to include BSM layers and vias as defined in [25] and [26] and shown in Fig. 2. SiO_2 was used as the back-side ILD, with thickness selected based on the BS vias (VB2–4). Considering predominant scaling trends in buried and BS interconnects, specialized standard cells EEQ were designed with pins on BSMs as shown in Fig. 3, to improve pin access and reduce route congestion during place and route. For example, the two-input and logic standard cell has the base variant named AND2, which has all pins on FS metals. The additional EEQ variants of this cell are AND2_BSM1, AND2_BSM2, and so on, in which various permutations of pins are moved to the BSMs. For BS power and signal routing, BSM1 employs a wider PG pitch (10 μm for NVDLA, Fig. 2) than the BS layers BSM2–4, leaving sufficient tracks for signal routing. Standard cells with BSM1 pins are placed using EEQ variants alongside PG routes on BSM1. The tighter pitch ($\leq 5 \mu\text{m}$) on BSM2–4 reduces static IR drop with BS power.

A. BS INTERCONNECT MODELING

The technology layout abstracts like library exchange format (LEF) files is updated to include BSMs and BVs. The interconnect technology file (ITF) used to generate NXTGRD for StarRC parasitic extraction contains the specification of BSMs (BSM1–4) and vias (VB2–4) and dielectric parameters capturing parasitic RC as well as crosstalk coupling capacitance as per Table 1. The higher aspect ratio lowers the resistance of metals, but via resistance increases due to increased contact resistance. The resistance values are scaled

TABLE 1. Key lithography assumptions for BS interconnects to enable clock and signal routing with BSMs and BVs.

Metal Layer	Width _{signal} (nm)	Width _{pwr} (nm)	Pitch _{min} (nm)	RPSQ ($\Omega/\square$)	Aspect Ratio
BSM1	24	140	84	2.22	1 : 2.8
BSM2	24	320	84	1.55	1 : 3.33
BSM3 & 4	38	680	98	1	1 : 3.33

Metal Layer	Top – Bottom layers config	Area (μm^2)	RPV (Ω/via)	Aspect Ratio
VB2	BSM1 – BSM2	0.000576	32	1:3
VB3	BSM2 – BSM3	0.000798	25	1:3
VB4	BSM3 – BSM4	0.00144	13.5	1:3
BSC [15] [25]	G, S/D – BSM1	0.000576	52.01	1:8

from [25] based on the aspect ratio and area considered. Similarly, the interconnect technology (ICT) file is used to generate Quantus RC (QRC) technology file for running QRC extraction during design implementation, as shown in Fig. 4.

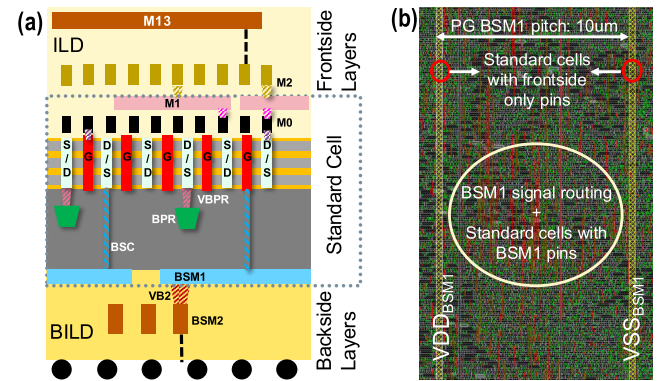


FIGURE 2. (a) Implementation of BSCs that enable direct gate and source/drain access for BSM routing within standard cells. This approach allows flexible placement of input/output pins on the FS (M1) and/or BS (BSM1), enabling the design of EEQ standard cells with multiple layout variants based on the selected pin-access configuration. (b) Signal, power routing on BSM1 in NVDLA.

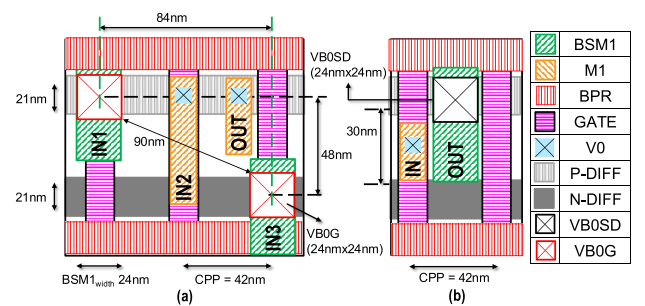


FIGURE 3. Pin access for (a) 3-input NAND and (b) inverter standard cell using FS and BS metals and contacts (VB0SD, VB0G). Tighter design-rule constraints, such as wider metals with increased pitch, limit the number of consecutive BS input pins (IN1, IN2, IN3) for the NAND3 cell. Providing multiple EEQ layouts with FS and/or BS pins enhances pin-access flexibility for place-and-route.

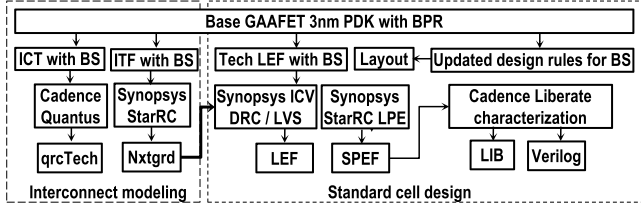


FIGURE 4. Design flow setup for interconnect modeling and standard-cell design to enable BS interconnects.

B. DESIGN RULES

Key lithography assumptions for BSMs are as follows. They are fabricated using Ruthenium or Tungsten [25] to withstand the high temperatures encountered during front-side BEOL (FSBEOL) processing, feature higher aspect ratios for reduced resistance, and are subject to tighter design rule constraints. When enabling BSMs for standard-cell routing, a large spacing requirement is introduced to account for overlay and misalignment errors associated with direct BS contact [8], [10], as illustrated in Fig. 3. The corner-to-corner (CtC) spacing for BSCs connecting source/drain to BSM1 (VB0SD) and gate to BSM1 (VB0G) is set at 90 nm. The horizontal spacing for BSM1 is defined as two contacted poly pitches (CPPs), totaling 84 nm, to facilitate pin access for alternate input terminals of standard cells. The tip-to-tip (TtT) spacing for the BSM1 layer is set at 48 nm, and the width of BSM1 matches that of the mid-BEOL metal stack width of FS metals, as shown in the following equation:

$$\text{BSM1}_{\text{width}} \in (\text{M5}_{\text{width}}, \text{M6}_{\text{width}}). \quad (1)$$

Fig. 3 shows example routing on BS Metal1 (BSM1) layer and corresponding standard-cell pin access points (gate terminals of a series transistor) for NAND3. As observed, for input pins, multiple routing options are possible based on front versus BSM access. All three input pins—IN1–IN3—cannot be placed on both FS and BS due to spacing constraints of BSMs. The number of FS metals (M0–M13) and vias (V0–V12) remains the same as base PDK for all evaluations.

C. STANDARD-CELL DESIGN

With standard-cell power rails moved to BS, cell height on both FS and BS could be reduced from 144 nm (5 M0-Track) to 96 nm (3 M0-Track) with BSC, BS signal route, and buried power rails as shown in Fig. 5. BPR width remains the same, but spacing between BPRs is reduced. Since input/output pins

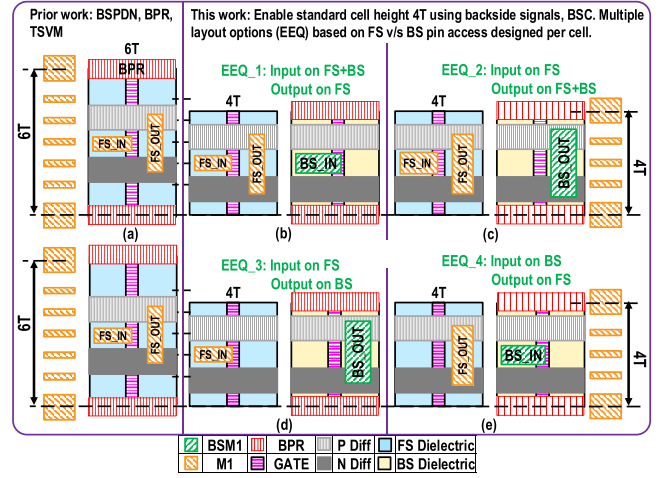


FIGURE 5. EEQ standard-cell layout abstract views with BS pins for an inverter. Transistor features remain unchanged. (a) Baseline with 5 M0-Track (6T), FS routing, and BPR. (b)–(e) FS and BS views with BS signal and BSC enable cell height scaled to 4T. Multiple pin access options are explored with 4T cells.

could be accessed from either side, multiple options for pin layers are made available while designing the standard-cell library. Tighter design rules for BSMs also limit the number of pin configurations per cell.

For this work, we have considered four EEQ options per cell as shown in Table 2. For simple standard cells with up to two inputs and a single output, all four EEQ options are easily realizable. However, for complex logic cells with three or more inputs, only options 1 and 2 are designed. For example, in a three-input (IN1–IN3) NAND gate, IN1 and IN3 are accessed using BSMs and IN2 and OUT accessed using FS metal as shown in Fig. 3. As observed, all three inputs and outputs cannot be accessed exclusively using front/BSMs only.

The standard-cell design flow (Fig. 4) involves updating 65 logic cells from a base 3-nm PDK to incorporate BSMs, with multiple layouts per cell (Fig. 5), resulting in 219 custom EEQ standard cells featuring FS and BS pins for enhanced place-and-route flexibility. BSM design rules are added to the DRC rule deck, and Synopsys IC Validator is used for DRC and LVS checks. After physical verification, Synopsys StarRC performs parasitic extraction to generate post-layout SPEF files for cell characterization. LEF are exported from ICV, and at 0.7 V and 25 °C, standard cells are characterized using RVT GAAFET models from [25] in Cadence Liberate, with standard-cell libraries and Verilog-A netlists generated post-characterization.

Three custom standard-cell libraries are developed using a baseline 3-nm GAAFET PDK to evaluate DTCCO experiments and design choices (Table 3). The baseline 6T library features an FS power delivery network (FSPDN) with power rails on buried interconnects (6T_FSBPR). By modeling BS interconnects, power delivery is enabled through four BSMs/vias (6T_BSBPR), representing the current research trends.

TABLE 2. EEQ options considered per cell. Base 3-nm PDK has pins exclusively on FS. The choice of pin access between FS versus BS is used to design four types of layout per cell.

Cell type	Input pin	Output pin	Purpose
EEQ_1	FS & BS	FS	Base PDK variant with BS pins
EEQ_2	FS	FS & BS	Base PDK variant with BS pins
EEQ_3	FS	BS	Enable front to back signal route
EEQ_4	BS	FS	Enable back to front signal route

Two additional libraries are introduced: 6T_BSCLK and 4T. The 6T_BSCLK library contains a subset of cells, such as inverters, buffers, D-flip-flops, and AND/OR clock gating cells with BS pins to support clock routing [16]. Clock routes incorporate various commonly used constraints such as double-width, double-spacing, ground shielding, and specific layer choices from BSM4 to M13.

To extend BSMs for signal routing, 4T cells are designed with 3-M0 tracks for signal paths. These cells have power rails on BPR and I/O pins on FS M1 and/or BS BSM1 layers. Each pin offers front and/or BS access, enabling a reduction in standard-cell height from 5-M0 to 3-M0 tracks. While simple cells (and, or, nand, INV, and BUFF) retain pins on both sides, more complex multi-input and multifunctional cells (nand3, and3, nor3, aoi, and oai) have pins exclusively on FS or BS metal layers.

IV. DESIGN SETUP AND METHODOLOGY

Design benchmarks for custom standard cells with BSCs and metals are based on a range of current industrial ASIC applications, as shown in Table 4. Power vectors, generated after RTL simulation using Synopsys VCS [27], are used to optimize the designs for specific application goals, such as dynamic and static power by selecting appropriate workloads. For instance, using vectors like Googlenet_conv2_3 × 3_int16 and Conv_8 × 8_fc_int16 allows tuning the NVDLA design to different power profiles, with smaller configurations activating less than half of the logic, leaving most of the design in standby mode in regular operation workloads.

The implementation of ASIC employs Cadence Genus and Innovus tools [28], covering a diverse set of designs such as high-performance compute, GPUs, communication, encryption, and memory controllers while maintaining a consistent placement utilization baseline (50%) as shown in Fig. 6. High-performance cores are set at 1 GHz, and GPU/general-purpose designs operate below 100 MHz.

For standard-cell height scaling, area and performance experiments are conducted separately; area scaling adjusts the bounding box, while performance scaling varies clock uncertainty. The design implementation uses updated technology and standard-cell LEF files with features like buried

TABLE 3. Standard-cell library variants to enable different DTCO experiments. While 6T_FSBPR and 6T_BSBPR represent current technology trends, 6T_BSCLK and 4T enable BSCLK and BS signal routing. For BSCLK, only clock cells are enabled with BS pins. 4T cells enable BS pins for all cells.

Parameter	6T_FSBPR (Baseline) [25]	6T_BSBPR [14] [15]	6T_BSCLK [16]	4T
BS power	X	✓	✓	✓
BS clock	X	X	✓	✓
BS signal	X	X	X	✓
Cell height (nm)	144	144	144	96
Library cells with BS Metals	None	None	INV / BUFF / DFF / Clock gaters (AND, OR)	All cell types

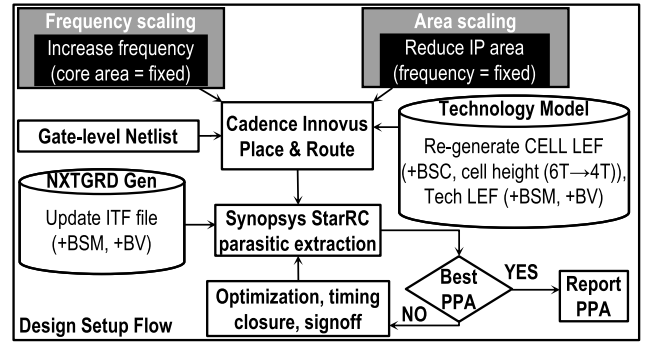


FIGURE 6. ASIC design flow is set up to evaluate standard-cell libraries and to study performance and area scaling trends across different IPs. Performance and area scaling trends are analyzed by varying clock uncertainty and design core area, respectively.

power rails and BSMs. An ITF file with BSM stack definition and dielectric parameters is used to generate an NXTGRD file incorporating parasitic and crosstalk modeling. DEF and NXTGRD files guide post-route parasitic extraction with Synopsys StarRC [29] to produce SPEF, with further optimizations performed to achieve the best PPA outcomes.

V. RESULTS

In this section, we evaluate the effectiveness of BS ICT at two abstraction levels. Section V-A presents standard cell-level improvements in timing, area, and power with BSMs/contacts, corresponding to the cell options described in Section III-C. We also examine delay and capacitance tradeoffs from the use of both FS and BSCs. Sections V-B and V-C show the area and performance scaling trends from cell height reduction across three benchmark categories: CPU (ibex, des_perf), GPU (NVDLA), and general-purpose SoC (dma_cntrl, wb_conmax, aes_core). Finally, Sections V-D and V-E present interconnect RC and IR-drop analysis for designs with 4T cells.

A. STANDARD-CELL DESIGN WITH BS METALS

In this section, we comprehensively analyze the benefits of enabling BSCs/metals with different custom standard-cell designs to enable BS power (6T_BSBPR), BSCLK (6T_BSCLK), and BS signal (4T). After comparing cell metrics, we systematically study the PPA benefits of enabling BS power, clock, and signal on CPU-representative ibex and GPU-representative system NVDLA.

Since enabling additional contacts on BS increases coupling capacitance, this requires pin capacitance, cell delay, transition times, and internal power to be updated. Transistor size remains unchanged across all types of standard-cell libraries and there is no change in cell leakage power while scaling down standard-cell height with 4T cells as shown in Table 5. Reducing standard-cell height by 2-M0 track results in reducing the area footprint by 33%. This enables compact placement of cells in physical design.

TABLE 4. Benchmark setup for different types of experiments—performance and area versus power scaling trends. Power vectors generated post-RTL simulation are used for physical design optimization. Designs are diverse in terms of cell versus wire dominance.

Benchmarks	Design (Part of IP cluster used in industry) ¹	Scaling trends studied	Configuration for power vector generation	Instance count
Ibex	RISC-V CPU core (CPU)	Performance, Area v/s Power	small	16k
NVDLA	Convolution & MAC (CMAC) unit of Nvidia Deep learning accelerator optimized for performance and dynamic power (GPU)	Performance, Area v/s Power	Googlenet_conv2_3x3_int16	1.45M
NVDLA_static	Convolution & MAC (CMAC) unit of Nvidia Deep learning accelerator optimized for area and static power (GPU)	Area v/s Power	Conv_8x8_fc_int16	1.45M
Wb_conmax	Wishbone interconnect IP (GP-SoC)	Area v/s Power	Vector-less	25k
Des_perf	DES encryption IP core optimized for performance (CPU)	Performance, Area v/s Power	Vector-less	60k
Dma_cntrl	Direct Memory Access controller (GP-SOC)	Area v/s Power	Vector-less	10k
Aes_core	AES Cipher security IP (GP-SOC)	Area v/s Power	Vector-less	12k

¹ Types of IP clusters - CPU: Central Processing Unit, GPU: Graphics Processing Unit, GP-SoC: Systems for General Purpose functions.

While enabling standard-cell library with BSCLK, the cell height remains unchanged to align with cells that would be used for power TAPs and combinational logic. This results in additional cap causing pin capacitance to increase 5%. However, the cell/transition delay and internal power indicate <1% degradation. 6T_BSCLK reserves BS interconnects exclusively for cells that would be used for clock propagation like inverters, buffers, clock gating cells, and leaf cells like flip-flops.

TABLE 5. Comparison of cell metrics for different inverter (INV_x1) layout architectures with drive strength of 1. The 6T_FSBPR and 6T_BSBPR configurations use similar standard cells with buried power rails. The 4T cells provide a 33% area reduction and a 10% decrease in pin capacitance, resulting in a 4% improvement in both cell delay and power.

Metrics for Cell:INV_x1	6T FSBPR	6T BSBPR	6T BSCLK	4T
Cell height (nm)	144	144	144	96
Cell width (nm)	84	84	84	84
Cpin (fF)	0.278	0.277	0.294	0.249
Lkg Pwr (pW)	164.7	164.7	164.7	164.8
Fast lib cell arc: input slew 2ps & output load 1fF				
Cell delay (ps)	12	11.9	12.1	11.5
Transition delay (ps)	14.18	14.15	14.29	13.6
Int. power (fJ)	0.052	0.051	0.0525	0.0499
Slow lib cell arc: input slew 28.5ps & output load 6.8fF				
Cell delay (ps)	74.5	74	75.1	71
Transition delay (ps)	89.8	89.8	90.5	86.2
Int. power (fJ)	0.0543	0.054	0.0548	0.0519

Extending the flexibility of using BS interconnects for any intracell routing or pin access enables a multitude of layout options per standard cell varying in front versus back pin access for every pin. In order to evaluate this, EEQ cells for each one of the 65 base standard-cell library were designed. This resulted in 219 new standard cells designed for a 4T cell library. Since BSM1 is wider compared to FS metals (M0–M4) and has a lower sheet resistance, the reduced effective RC improves cell and transition delay by 4%–4.5% with reduced height standard cells. Reduced pin cap also reduces cell internal power by 4% as shown in Table 5.

BSM routing reduces total wirelength. Fig. 7 shows clock routes with two types of configuration—baseline with FS clock

routing and enabling BS clock routing. As observed, clock trunk routing is done with wider and thicker BS metals with reduced sheet resistance improving clock slew and timing. For leaf routes, the physical design tool has an option of using either BS or narrower FS metals. Since the clock buffers, gating cells, and flip-flops are available with BS pins, clock leaf cells are routed using metal layers from both sides. This helps reduce clock tree cell count as shown in Fig. 8. Timing-driven designs like Ibex CPU leverage this opportunity most with reduced clock tree cell count and clock cell area. Clock timing improves with more routing resources for clock trunk and leaf nets close to silicon. Such a saving may not be realizable with adding more layers toward FS routing only designs due to increased proximity of access from silicon.

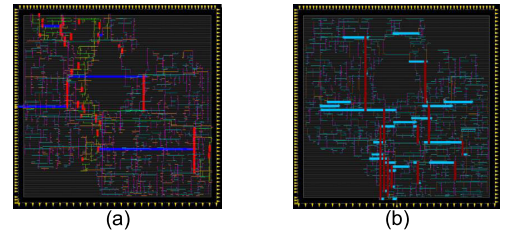
**FIGURE 7. Enabling clock routing on BS reduces clock wirelength. Clock trunk is routed on BS with wider metals BSM4/BSM3, while the remaining BS and FS metals are used for leaf routing. (a) Baseline with FS clock routing. (b) BSCLK routing.**

Fig. 8 shows PPA evaluation with different standard-cell libraries across two representative IPs Ibex and NVDLA. Ibex is performance and dynamic power driven, while NVDLA is area and static power dominant. With BSCLK routing, timing and performance improve by 12% for Ibex. Enabling BSMs for all cells improves performance further due to reduced average device size and improved placement and routing congestion. Design area is reduced most for NVDLA due to a greater opportunity to reduce device size with relaxed timing constraints compared to CPU-type

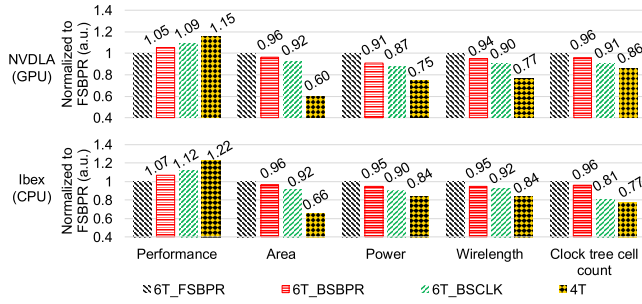


FIGURE 8. PPA analysis of ibex RISC-V CPU core and NVDLA with different standard-cell library configurations. 4T cells improves the performance by 10%–15% compared to the 6T_BSBPR baseline. Standard-cell area reduces by 30%–35% with shorter height cells. Total power also reduces by 10%–15% compared to BSBPR.

designs. While dynamic power is a major component for Ibex (>90%), static and dynamic power both contribute (25% and 75%) to the total power for NVDLA. Thus, low-power design methodologies leverage BS interconnects to reduce interconnect length, reducing net switching power. This also reduces average device size and thus lowers static power dissipation.

B. AREA SCALING

Reduced height standard cells enable area scaling directly due to their smaller cell area footprint. For area scaling experiments, the design area is reduced from 0% to 40%. Considering current ASIC design utilization trends targeted by industry for CPU/GPU/SoCs, the starting placement utilization target is set to 50%. Thus, the floorplan with no area scaling at 0% has a density of 50%. Design area reduction involves reducing floorplan bounding box and moving design ports without reordering.

With BS interconnects, routing congestion reduces at tighter area constraints as observed in Fig. 9. For large IPs like NVDLA, routing congestion is more pronounced due to higher leaf cell count post-RTL. With limited track availability close to silicon, we observe increased routing congestion

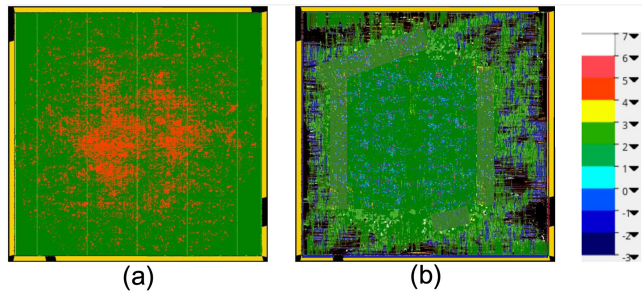


FIGURE 9. Routing congestion for FS metal layers M1–M5 for NVDLA. (a) FS routing shows congestion hotspots (red) spread across the design due to reduced track availability. (b) Enabling BS routing enables additional tracks close to silicon reducing routing congestion and wire length.

on M1–M5 as shown in Fig. 9. However, enabling BSMs reduces congestion and improves track availability for FS metal routing.

Fig. 10 shows area scaling trends for different design types based on PPA targets being optimized. The baseline for studying area scaling trends is using a standard-cell library with BSBPR. This aligns the baseline with the current research trend for sub-3-nm technology space. Reducing standard-cell height by 2-M0 tracks reduces cell area footprint by 33%. Timing-constrained designs like Ibex and des_perf achieve a maximum design standard-cell area savings of 30% while reducing power consumption by 10%.

The majority of power savings is from reduced internal and switching power. Internal power reduces due to the reduced average device size. Switching power reduces with reduced interconnect length and also device size reduction. NVDLA, aes_core, dma_cntrl, wb_conmax sees maximum design area savings of 35% while achieving $\geq 10\%$ power savings. NVDLA_static, which is optimized for static power dissipation also achieves 35% area reduction with 13% power savings. Although 4T standard cells have similar leakage profile compared to 6T cells due to similar device size, enabling pin access on both sides of wafer reduces the average device size of the design, which reduces static power.

C. PERFORMANCE SCALING

Performance scaling trend is studied by reducing clock frequency while keeping the design floorplan bounding box fixed. Since Ibex, NVDLA, and des_perf are dynamic power and timing constrained, performance scaling is best studied with these designs. Major factors driving performance scaling for modern-day processors are interconnect RC and device size (faster device with smaller footprint).

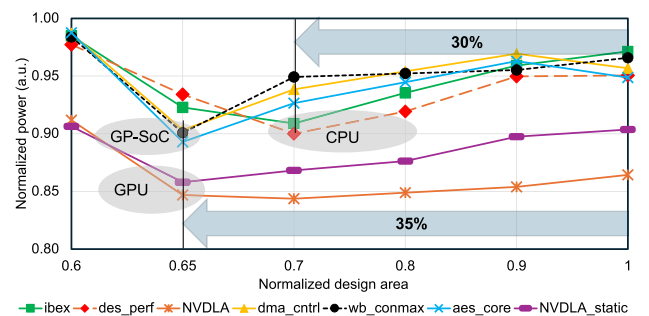


FIGURE 10. Area scaling trends for different types of designs. With standard-cell height reduced by 2-M0 tracks, the cell area is reduced by 33%. The core area of HPC designs reduces by 30%. General-purpose SoC and GPU-representative designs observe a max area reduction of 35% due to relaxed design constraints.

Performance scaling trends with 4T cells for HPC and GPU-representative designs are shown in Fig. 11. As the clock frequency increases, the dynamic and static power dissipation increases as per $P = CV^2f$ where P = power, C = capacitance, V = voltage, and f = frequency. However,

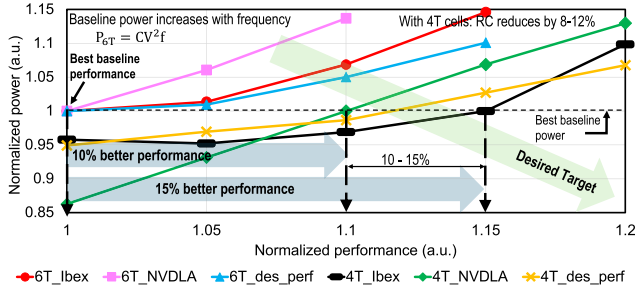


FIGURE 11. Performance scaling trend with 4T cells for Ibex, NVDLA, and des_perf with increasing clock frequency. The 10%–15% performance improvement is observed with BS interconnects.

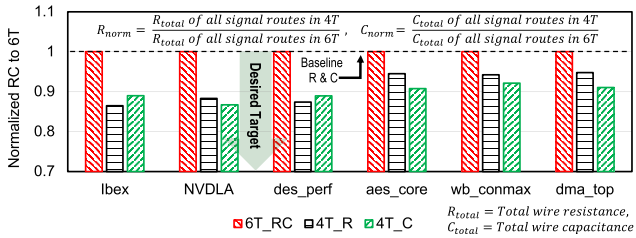


FIGURE 12. Interconnect RC trend with BSMs for all design types. Average standard-cell output pin cap reduces for performance-driven designs. With BS metals and 4T cells with exclusive BS pin access, cross-cap between metals close to silicon reduces. Cell pin cap reduces with 4T cells and reduces further while optimizing for timing.

with advanced nodes, the interconnect RC also contributes to path timing significantly at tighter frequencies. With BS metals, the interconnect wire length and resistance are reduced. The 4T cells also have smaller C_{pin} compared to 6T cells as per Table 5. This reduces dynamic power as well as improves path timing. Timing versus power tradeoff results in performance improvement in the range of 10%–15% for different types of designs.

D. RC TRENDS WITH BS METALS

BSM adds layers to the processing metal stack. Compared to adding metals for an FS routing only BEOL, BSMs also enable direct access to transistors from BS. Thus, BS metals provides additional layers close to silicon affecting standard-cell sizing and interconnect RC for signal/clock routing.

Standard-cell pin cap and interconnect RC trends for different designs are shown in Fig. 12. Average standard-cell output pin cap reduces for performance-driven designs. Constrained on timing, such designs also observe an improvement in interconnect resistance, mainly from reduced wirelength with 4T cells and shorter pin-to-pin routing with BS metals. For low-power designs, cell pin cap and wire cap are dominant metrics tracked for optimization. With BS metals and 4T cells with exclusive BS pin access, cross-cap between metals close to silicon reduces. Cell pin cap reduces in 4T cells while optimizing for timing.

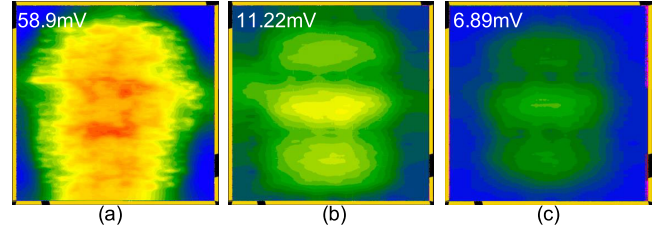


FIGURE 13. Static IR-drop analysis for NVDLA design. (a) IR drop with 6T cells and FS routing. (b) IR drop with 6T cells and BS power. (c) IR drop with 4T cells and BS power, clock, and signal routing.

E. IR-DROP ANALYSIS

In a sub-3-nm technology node, interconnect resistance increases drastically due to reduced metal widths and aggressively scaled metal pitches, resulting in power integrity issues [15]. With FS routing, a critical interplay between power and clock, the signal route becomes crucial while resolving static IR-drop issues without degrading PPA. We perform IR-drop analysis using Cadence Voltus Power IC integrity on the placed and routed NVDLA design. Fig. 13(a) shows >5% IR drop observed with NVDLA operating at 700 mV. Increased pin access and route congestion with higher cell density at the design core cause hotspots. Increasing power grid density causes PPA degradation due to the criticality of timing paths in this region. From Fig. 13(b), with BS power routing, congestion reduces and denser power grid leveraging least resistive BSMs reduces static IR drop to <5%. With BS signaling, instance IR drop reduces with reduced switching and internal power from 4T cells. Static IR drop continues to be <1% with improved timing as shown in Fig. 13(c). Worst case IR drop reduces by 80% with BS power. Enabling BS signaling with 4T cells reduces IR drop further by 32%. While BSM helps with power route, shorter stub routes help signal routing and improve overall PPA.

VI. CONCLUSION

This work demonstrates that enabling BSCs and signal interconnects creates a significant DTCO opportunity for standard-cell height reduction and BS interconnect optimization in advanced nodes. By allowing direct connection of gate, source/drain terminals to BSMs, our approach facilitates design of EEQ standard cells with flexible input/output pin placement on FS and BS, enabling 2-M0 track reduction in 3-nm nanosheet technology.

Comprehensive experimental results across a range of industrial representative designs show that 4T cells achieve up to 35% area reduction and 10%–15% total power savings. For high-performance designs, a 15% increase in max performance is realized at iso-power compared to BSBPR configurations. These improvements are attributed to an 8%–12% reduction in device and interconnect RC, as well as lower pin capacitance, which altogether enable up to 32% improvement in static IR drop. Future work must address thermal–electrical co-simulation to quantify

thermal versus PPA tradeoffs across BS configurations, as BS routing introduces complex multidirectional heat flow scenarios requiring comprehensive device-to-package thermal modeling.

Overall, enabling BS interconnects for standard cell, clock, and signal routing extends standard-cell scaling while preserving critical transistor geometries. Our results highlight that BS routing for signals and clock—not just power delivery—represents the next phase of technological innovation, unlocking generational PPA improvements for sub-3-nm technology nodes.

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